

The Transformer Architecture

Overview

The Transformer, introduced by Vaswani et al. in 2017 in the paper 'Attention Is All You Need', is a neural network architecture that relies entirely on attention mechanisms, dispensing with recurrence and convolutions. It has become the dominant architecture in natural language processing and has spread to vision, audio, and multimodal tasks.

Prior sequence models like RNNs and LSTMs process tokens sequentially, making parallelization difficult and long-range dependency learning challenging. The Transformer solves both problems by processing all tokens simultaneously and connecting any two positions with a constant number of operations.

Encoder-Decoder Structure

The original Transformer follows an encoder-decoder design. The encoder maps an input sequence of tokens to a sequence of continuous representations. The decoder takes those representations and generates an output sequence one token at a time, attending to both its own previous outputs and the encoder output.

Each encoder layer consists of two sub-layers: multi-head self-attention and a position-wise feed-forward network. Each decoder layer adds a third sub-layer: cross-attention over the encoder output. Residual connections and layer normalization are applied around each sub-layer.

Positional Encoding

Because the Transformer processes all tokens in parallel with no inherent notion of order, positional information must be injected explicitly. The original paper uses sinusoidal positional encodings: fixed vectors added to the token embeddings before the first layer.

The encoding for position pos and dimension i uses $\sin(pos / 10000^{(2i/d_{model})})$ for even dimensions and $\cos$ for odd dimensions. This choice allows the model to generalize to sequence lengths not seen during training.

Later models like BERT use learned positional embeddings, which are trained end-to-end. Relative positional encodings (as in T5 and RoPE) have become popular as they generalize

better to long sequences.

Feed-Forward Sub-Layer

Each Transformer layer contains a position-wise feed-forward network (FFN) applied independently to each position. It consists of two linear transformations with a ReLU (or GELU in later models) activation in between: $\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2$.

The inner dimension of the FFN is typically four times the model dimension. For a model with $d_{\text{model}}=512$, the FFN inner dimension would be 2048. This sub-layer is responsible for much of the model's capacity and has been shown to store factual knowledge.

Residual Connections and Layer Normalization

Each sub-layer (attention and FFN) is wrapped with a residual connection followed by layer normalization: $\text{output} = \text{LayerNorm}(x + \text{Sublayer}(x))$. Residual connections ease gradient flow during training, allowing very deep networks to be trained effectively.

Layer normalization normalizes across the feature dimension for each token independently, making training more stable. Some models use pre-norm (normalizing before the sub-layer) rather than post-norm, which has been found to improve training stability for very deep models.

Encoder-Only and Decoder-Only Variants

Many modern models use only part of the original encoder-decoder architecture. BERT is encoder-only: it processes the full input bidirectionally and is used for understanding tasks like classification and question answering.

GPT and its successors are decoder-only: they use masked self-attention so each token only attends to previous tokens, making them suitable for autoregressive text generation.

T5 and BART retain the full encoder-decoder structure and are used for sequence-to-sequence tasks like summarization and translation.

Scaling and Efficiency

Transformers scale remarkably well with data and parameters. Scaling laws (Kaplan et al., 2020) show that model performance improves predictably as a power law with compute, data, and parameters.

The main computational bottleneck is the $O(n^2)$ complexity of self-attention with respect to sequence length. Research into efficient attention variants — sparse attention, linear attention, and sliding window attention — aims to reduce this cost for long sequences.

Models like Longformer, BigBird, and FlashAttention address this bottleneck through algorithmic improvements, enabling Transformers to handle documents, code repositories, and long-form content.